

# **Workflow Management mit Nextflow**

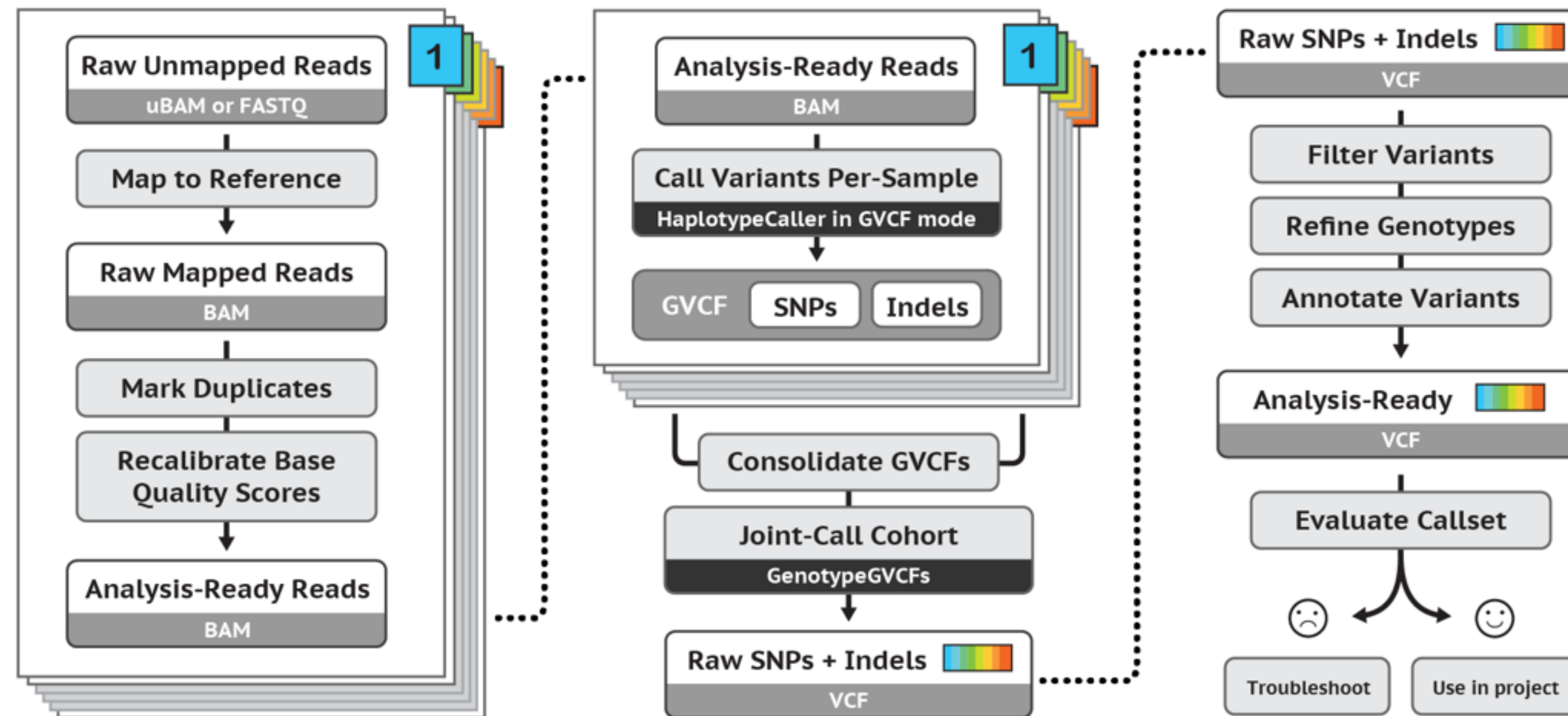
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# Typical bioinformatics pipelines

- Although the conceptual analysis pipeline has only a few steps, in practice the implementation consists out of a significant amount of sequential processing steps
- Each step can consist for example of running an already existing piece of software or integrating a newly developed script
- Sometimes a step is “just” parsing the data from one format to another ( for example: .sam > .bam > sorted.bam)

# Variant Calling Workflow

## GATK Best practices



*Best Practices for SNP and Indel discovery in germline DNA  
- leveraging groundbreaking methods for combined power  
and scalability.*

Source: <https://gatk.broadinstitute.org>

# NGS Projects

- 1000 Genome Project
- Genome Aggregation Database/ Exome Aggregation Consortium
- 1000 Plant Transcriptomes Initiative (1KP)
- Darwin Tree of Life

-> **Need for reproducibility and scalability!**

# Reproducibility

- Fully reproducible analysis pipelines are challenging
- Differences in hardware, operating systems and software versions
- Best practices are being established, however their actual implementation is still on case-by-case basis

# FAIR principles

- FAIR Guiding Principles for scientific data management
  - **F**indable
  - **A**ccessible
  - **I**nteroperable
  - **R**eusable
- <https://www.go-fair.org/fair-principles/>

# Workflow management systems

- Nextflow
- snakemake
- Galaxy
- Cromwell
- and there are more...

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# What is Nextflow?

- **DSL (Domain Specific Language)** for parallel and scalable computational pipelines
- enables **reproducibility** through Docker and Singularity container support
  - you can write self-contained pipelines, manage versions and rapidly reproduce any former configuration
- **portable** on multiple platforms through abstraction layer between the pipeline's logic and execution layer
  - locally, HPC, cloud

# What is Nextflow?

- Based on the **dataflow** programming paradigm
- Tasks are defined as **processes** (isolated units of computation written in any scripting language)
- Processes communicate with each other through **channels** (asynchronous FIFO queues that can send and receive any kind of data)

# Simple example

```
process sayHello{
  input:
    | val cheers
  output:
    | stdout
  """"
  echo $cheers
  """"
}

workflow{
  | channel.of('Ciao','Helo','Hola','Salut') | sayHello | view
}
```

# Advantages

- easy integration of existing scripts and tools in a Nextflow pipeline
- tracks all the intermediate results produced during the pipeline execution and allows you to resume its execution from the last successfully executed step
- flexible execution:
  - **HPC workload managers:** GridEngine, SLURM, LSF, Moab, etc.
  - **cloud services:** AWS (Amazon Web Services), Google Cloud, Microsoft Azure
  - **Kubernetes** (open-source system for automating the deployment, scaling, and management of containerized applications)

# Nextflow Ressources

- Documentation:

<https://www.nextflow.io/docs/latest/index.html>

- Learn Nextflow:

<https://www.nextflow.io/blog/2023/learn-nextflow-in-2023.html>

- Nextflow Patterns:

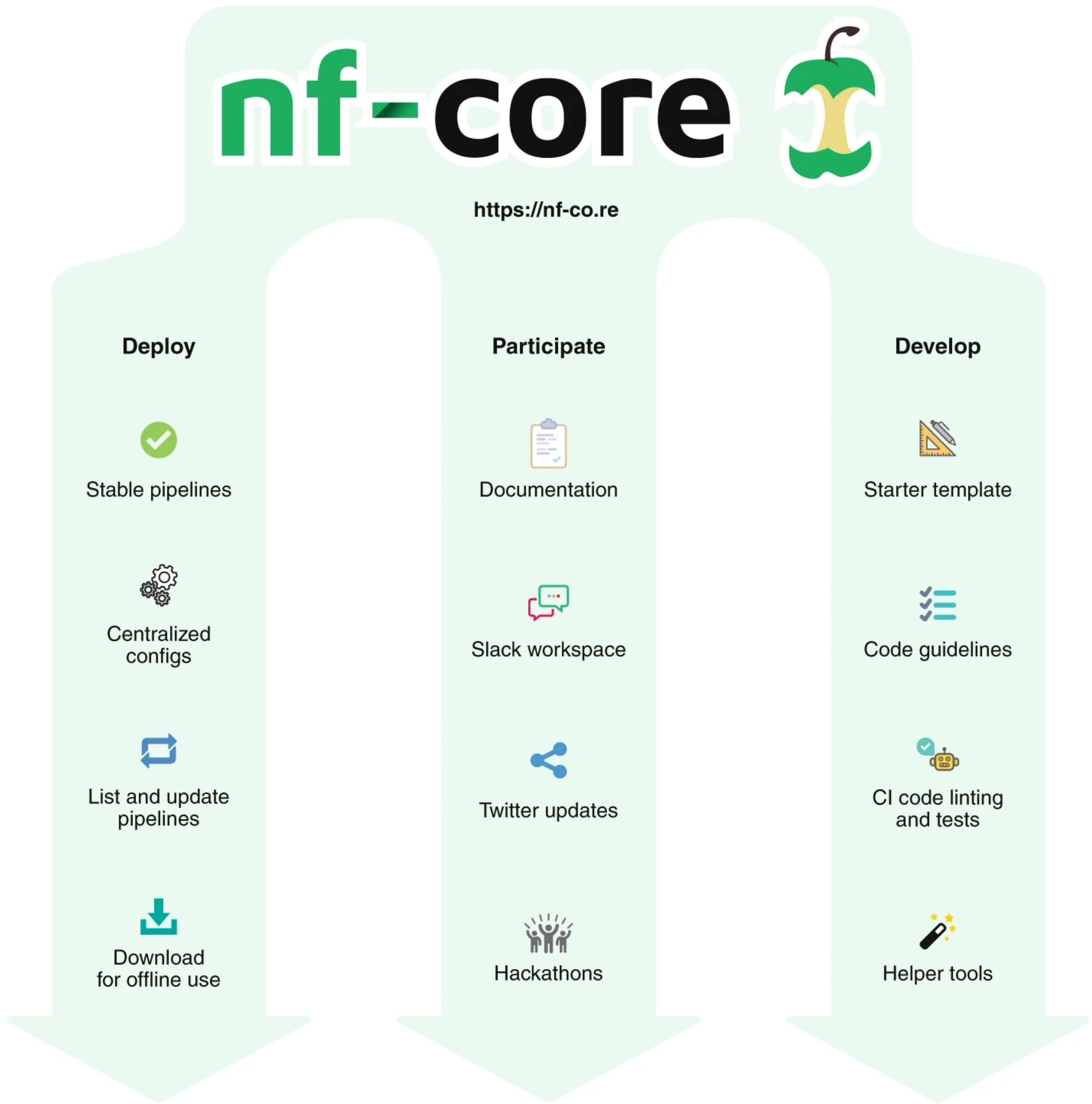
<https://nextflow-io.github.io/patterns/>

# nf-core

- community effort to collect a curated set of analysis pipelines built using Nextflow: **<https://nf-co.re/pipelines>**
- nf-core tools is a python package for helping with developing nf-core pipelines.
- options for listing, creating, and even downloading pipelines for offline usage
- easy installation via conda

```
conda install nf-core
```

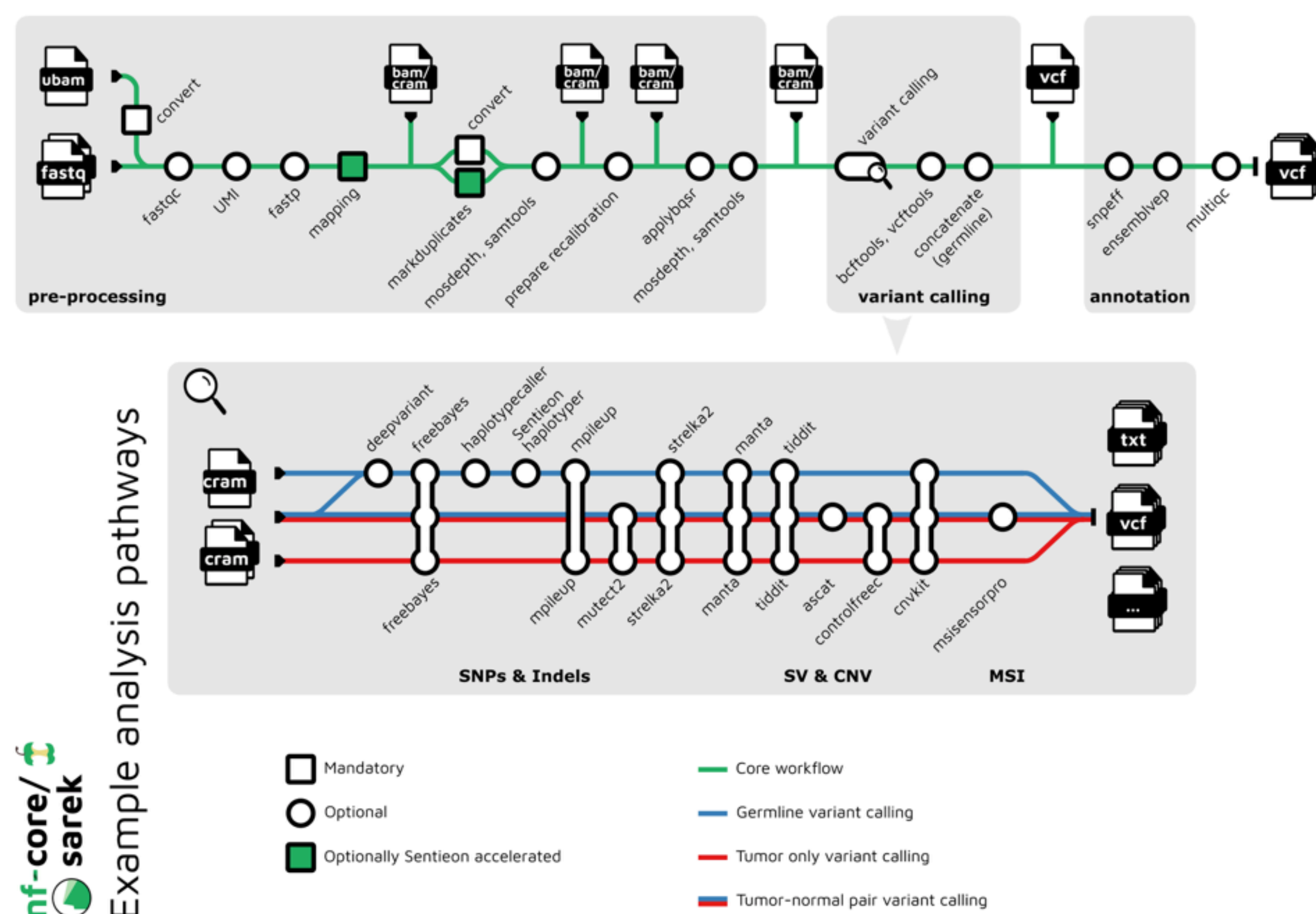
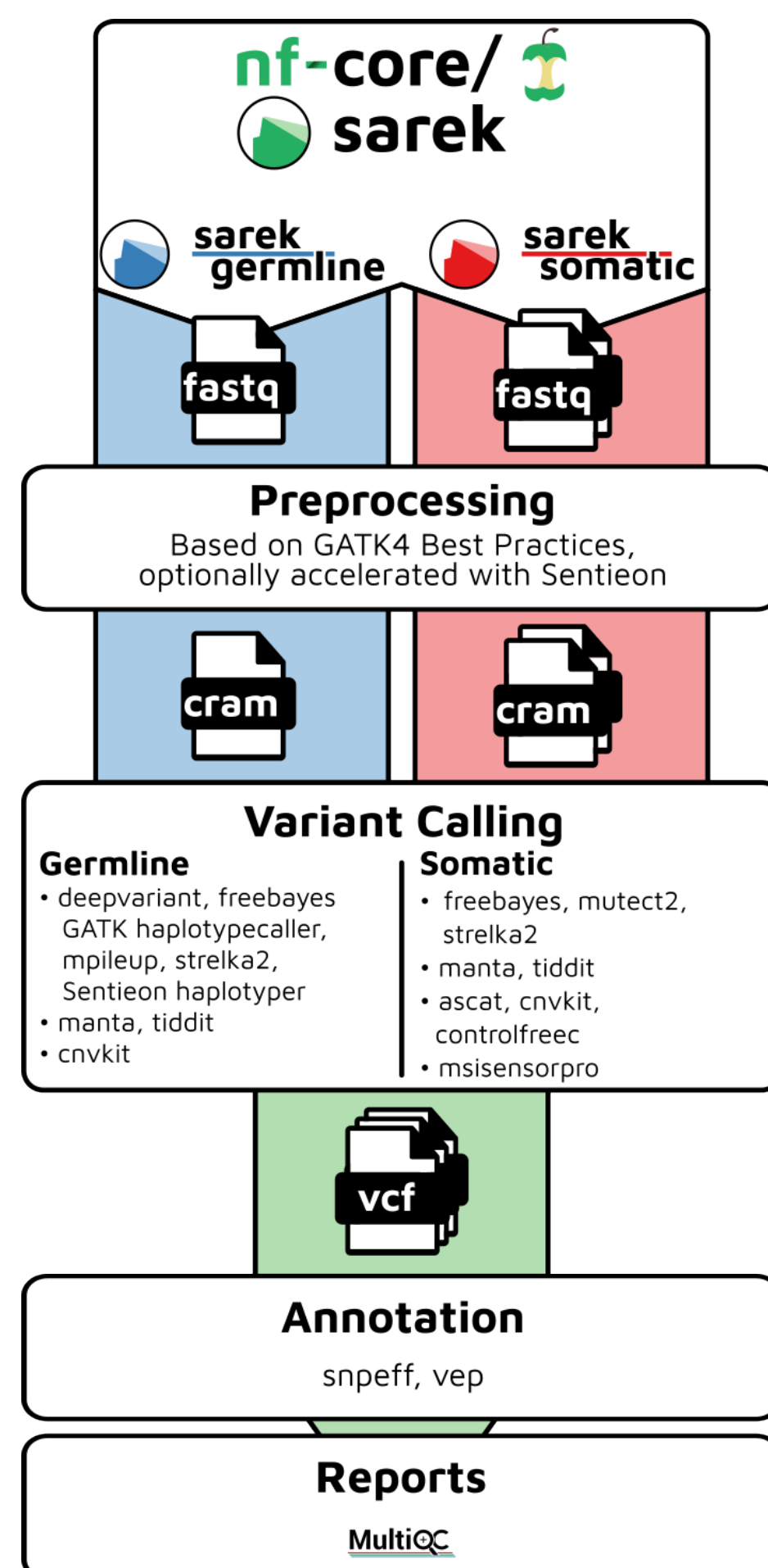
# nf-core



Ewels, P.A., Peltzer, A., Fillinger, S. *et al.* The nf-core framework for community-curated bioinformatics pipelines. *Nat Biotechnol* **38**, 276–278 (2020). <https://doi.org/10.1038/s41587-020-0439-x>

# nf-core/sarek

Analysis pipeline to detect germline or somatic variants (pre-processing, variant calling and annotation) from WGS / targeted sequencing



# nf-core/differentialabundance

Differential abundance analysis for feature/ observation matrices from platforms such as RNA-seq

